

Cloud Computing for AI

SITAMS CSE (AI & ML) • R23 • III Year • V Semester • Course Code: 23CAI351T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit I: INTRODUCTION TO CLOUD COMPUTING AND AI INTEGRATION

Syllabus topics

- Basics of Cloud Computing: Characteristics, Models, and Services, Cloud Service Models: IaaS, PaaS, SaaS, Deployment Models: Public, Private, Hybrid, Community, AI and Cloud Convergence: Benefits and Challenges, Use Cases of AI in Cloud: NLP, Vision, Analytics, Overview of Cloud Providers for AI: AWS, Azure, GCP

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

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Unit II: STORAGE, COMPUTING, AND DATA PROCESSING IN THE CLOUD

Syllabus topics

- Cloud Storage Services: S3, Blob, BigQuery, Virtualization and Elastic Computing, Distributed Computing with Hadoop and Spark, Data Ingestion and Processing Pipelines, Data Lakes and Warehousing in the Cloud, Cost Optimization for Storage and Compute Resources

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Unit III: CLOUD-BASED MACHINE LEARNING AND DEEP LEARNING

Syllabus topics

- ML Services on AWS (SageMaker), Azure ML, GCP Vertex AI, Training and Deploying Models on Cloud, AutoML and Custom ML Model Workflows, GPUs/TPUs for Model Training, Experiment Tracking and Model Evaluation, Integration of Notebooks (Jupyter, Colab) with Cloud Storage

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Unit 4: ADVANCED CLOUD CONCEPTS FOR AI APPLICATIONS

Syllabus topics

- Containers and Docker for AI Applications, Kubernetes and Cloud-native AI Workflows, Serverless Computing: AWS Lambda, Azure Functions, CI/CD Pipelines for AI Models in Cloud, Scaling AI Applications using Load Balancers and Auto-Scaling
- Monitoring and Logging in Cloud for AI Workflows

Revision checklist

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Unit 5: SECURITY, ETHICS, AND CASE STUDIES IN CLOUD AI

Syllabus topics

- Security and Privacy in Cloud-based AI, Identity and Access Management (IAM) in Cloud, Cost Management and Billing for AI Services, Ethical Issues and Fairness in Cloud AI, Case Study: AI in Healthcare Cloud Solutions, Case Study: Real-Time Analytics in Financial Cloud Services

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Study Hub status

This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.